Route Requests Through Phoenix

As requests come into an application, it's important that they get sent to the right function in your application. Along the way, they'll get transformed and interacted with by middleware functions. Middleware can serve any purpose, but you'll often write middleware to set common data and validate that the request can be served.

Phoenix uses the Plug[12] library, which refers to middleware as "plugs." Plugs can be either functions or modules.

Let's look at how Phoenix encapsulates plugs inside of the Phoenix.Endpoint module. Once we get these basics down, we'll move on to our controller.

How Requests Flow

The Phoenix generator creates several files as part of its generation process. These files relate to one of three things: your application code that lives in PhoneApp, your web-specific code that lives in PhoneAppWeb, or the configuration files. The App and AppWeb namespaces are best practices that encourage separation and clarity between your application logic and web logic.

One of the files that the Phoenix generator created—when you ran mix phx.new in Chapter 6—is called PhoneAppWeb.Endpoint. An Endpoint[13] manages the request lifecycle for any requests sent to it. (By default, your application has a single endpoint, but you can actually mount multiple endpoints on different ports.)

Our PhoneAppWeb.Endpoint module found at lib/phone_app_web/endpoint.ex has about 50 lines of code that outline exactly what happens in our application request lifecycle. The basic shell looks like this:

phone_app/lib/phone_app_web/endpoint.ex

```
defmodule PhoneAppWeb.Endpoint do
  use Phoenix.Endpoint, otp_app: :phone_app

  # ... Many lines outlining your application's request flow
end
```

The line use Phoenix.Endpoint brings in all of the functionality needed for this module to be an endpoint. This file has many more lines, but we won't go through each one. Instead, here's the most important section:

12. https://hexdocs.pm/plug/Plug.html
13. https://hexdocs.pm/phoenix/Phoenix.Endpoint.html